
(a)
$U = 0$: $\nu_k \in \{0, 1\}$, $S_{\text{corr}} = 0$
1 1 1 1 0 0 0 0
(b)
$U = 6t$: $0 < \nu_k < 1$, $S_{\text{corr}} = 4.22$
0.82 0.82 0.73 0.73 0.27 0.27 0.18 0.18